

Mark-to-Market (MTM) Settlement

Module: Futures & Options

Chapter: 04 - Mark-to-Market Settlement

Level: Beginner → Intermediate

Learning Objectives

After completing this chapter, you will be able to:

- Understand Mark-to-Market settlement.
 - Explain daily profit and loss adjustment.
 - Calculate MTM for long and short futures positions.
 - Understand margin requirements.
 - Learn the role of clearing corporations.
 - Differentiate between daily settlement and expiry settlement.
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Introduction

In futures trading, contracts are not settled only at expiry.

Instead, every trading day:

- The contract value is recalculated.
- Profits are credited.
- Losses are debited.

This daily adjustment process is called **Mark-to-Market (MTM)**.

Definition

Mark-to-Market is the daily process of calculating and settling profit or loss on futures contracts based on the closing market price.

Why MTM is Required?

Without MTM:

- Losses could accumulate for a long period.
 - Traders may default on payments.
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- Counterparty risk increases.

MTM ensures:

- Daily risk control.
- Faster settlement.
- Market stability.

Futures Settlement Flow

flowchart TD

A["Open Futures Position"]

--> B["Day End Closing Price"]

B --> C["Compare With Previous Price"]

C --> D["Price Movement"]

D -->|Price Goes Up| E["Long Position Profit"]

D -->|Price Goes Down| F["Long Position Loss"]

D -->|Price Goes Down| G["Short Position Profit"]

D -->|Price Goes Up| H["Short Position Loss"]

E --> I["Credit Margin Account"]

F --> J["Debit Margin Account"]

G --> I

H --> J

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How MTM Works

Every trading day:

Opening Futures Price



Closing Market Price



Calculate Difference



Profit Added / Loss Deducted



Updated Margin Balance

MTM Formula

Long Position (Buyer)

A buyer profits when futures prices increase.

Formula:

Profit/Loss =
(Current Price - Previous Price) × Lot Size

Short Position (Seller)

A seller profits when futures prices decrease.

Formula:

Profit/Loss =
(Previous Price - Current Price) × Lot Size

Example 1: Long Futures Position

A trader buys Nifty Futures.

Details	Value
Buy Price	₹22,000
Lot Size	50
Closing Price	₹22,200

Calculation:

$$\begin{aligned} & (22,200 - 22,000) \times 50 \\ &= 200 \times 50 \\ &= ₹10,000 \end{aligned}$$

Result:

$$\text{Profit} = ₹10,000$$

The amount is credited to the trader's margin account.

Example 2: Long Position Loss

Buy Price:

₹22,000

Next day closing price:

₹21,800

Lot Size:

50

Calculation:

$$(21,800 - 22,000) \times 50$$

$$= -200 \times 50$$

$$= -\text{₹}10,000$$

Result:

$$\text{Loss} = \text{₹}10,000$$

The amount is deducted from margin.

Example 3: Short Futures Position

A trader sells futures.

Details	Value
Sell Price	₹1,000
Lot Size	100

Price falls to:

₹950

Calculation:

$$(1,000 - 950) \times 100$$

$$= 50 \times 100$$

$$= \text{₹}5,000$$

Result:

$$\text{Profit} = \text{₹}5,000$$

MTM Daily Cycle

flowchart LR

A["Position Opened"]

--> B["Closing Price Calculated"]

B --> C["Daily Profit/Loss"]

C --> D{"Result"}

D -->|Profit| E["Credit Margin Account"]

D -->|Loss| F["Debit Margin Account"]

E --> G["Continue Position"]

F --> G

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Margin Account

When a futures position is opened:

Trader deposits Initial Margin

Daily:

Result	Action
Profit	Added to account
Loss	Deducted from account

Margin Call

Meaning

A margin call occurs when the account balance falls below the required margin level.

The trader must add additional funds.

Example

Required Margin:

₹50,000

Available Balance:

₹40,000

Shortfall:

₹50,000 - ₹40,000

= ₹10,000

Trader must deposit:

₹10,000

Role of Clearing Corporation

The clearing corporation acts as an intermediary between buyers and sellers.

Responsibilities:

- Calculates MTM obligations.
 - Transfers profits and losses.
 - Maintains settlement guarantee.
 - Reduces default risk.
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MTM vs Final Settlement

Feature	MTM Settlement	Final Settlement
Timing	Daily	Expiry
Purpose	Risk management	Contract completion
Frequency	Every trading day	Once
Price Used	Closing price	Final settlement price

MTM Example Timeline

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timeline
    title Futures MTM Timeline
    Day 1 : Buy Futures @ ₹1000
           : Deposit Margin
    Day 2 : Closing Price ₹1050
           : Receive ₹50 Profit
    Day 3 : Closing Price ₹1020
           : Pay ₹30 Loss
    Expiry : Final Settlement

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Advantages of MTM

Risk Reduction

Losses are settled immediately.

Transparency

Daily prices reflect market reality.

Safety

Reduces chances of default.

Liquidity

Allows continuous trading.

Risks of MTM

1. Daily Cash Requirement

Losses must be paid immediately.

2. Margin Pressure

Large price movements can trigger margin calls.

3. Forced Liquidation

Failure to maintain margin may close positions.

Real-Life Hedging Example

An airline buys fuel futures.

If crude oil prices rise:

- Fuel cost increases.
- Futures position gains value.

The MTM profit helps offset higher fuel expenses.

Key Terms

Term	Meaning
MTM	Daily profit/loss settlement
Closing Price	End-of-day market price
Margin	Security deposit
Margin Call	Additional funds required
Clearing Corporation	Settlement authority
Long Position	Buyer of futures
Short Position	Seller of futures

Summary

- MTM is a daily settlement process.

- Futures profits and losses are adjusted every day.
- Long traders benefit from rising prices.
- Short traders benefit from falling prices.
- Margin accounts are updated daily.
- MTM reduces market risk.

Quick Revision

- ✓ MTM = Daily settlement
- ✓ Profit → Credit
- ✓ Loss → Debit
- ✓ Long → Profit when price rises
- ✓ Short → Profit when price falls
- ✓ Margin Call → Additional deposit
- ✓ Clearing Corporation → Settlement safety

Practice Questions

Concept Questions

1. What is Mark-to-Market settlement?
2. Why is MTM important in futures markets?
3. Explain the role of margin accounts.

Numerical Questions

Question 1

A trader buys futures at ₹5,000.

Lot size = 200

Closing price = ₹5,050

Calculate MTM profit.

Question 2

A trader sells futures at ₹10,000.

Lot size = 100

Price falls to ₹9,800.

Calculate profit.

What's Next?

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Next chapter:

- Spot price
- Futures price
- Cost of carry model
- Basis
- Premium and discount
- Factors affecting futures prices